

Grab System

1.1.5

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Chapter 1

Namespace Index

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2.1 Class Hierarchy

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Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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The base component for all grabbers that implement the ' ConditionalGrab() ' method. Also implements repeat grab attempt routine functionality	14
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A simple component that references a IgnoreCollidersWhileGrabbing component that uses the 'OverrideUnignoreDelegate' delegate event to delay the unignore collisions for some time	17
GrabSystem.DistanceCatchAssist	
A component that is generally NOT attached to the same GameObject as a DistanceGrabber that must be referenced to enable assisted catching. This is NOT attached to the same GameObject as a DistanceGrabber to avoid triggering unwanted components	18
GrabSystem.DistanceGrabbable	
A component that can be attached to the same GameObject as a GrabbableObject that allows a Grabber to instantly grab it from a distance.	18
GrabSystem.DistanceGrabber	
A component that is attached to the same GameObject as a Grabber to enable it to grab DistanceGrabbable objects from a distance.	23
GrabSystem.Poses.MaintainOffsetByGrabOffset.Entry	25
GrabSystem.Poses.MaintainOffsetByProjectedAngle.Entry	26
GrabSystem.Structures.FloatMinMax	26
GrabSystem.ForceDistanceGrabbable	
A component that is derived from ' DistanceGrabbable ' that allows for force based grabbing of a GrabbableObject that also has a Rigidbody component. Modes: Instant - Teleports the object to the grabber and grabs it instantly (as a normal DistanceGrabbable would). Arch - Uses physics forces to pull the GameObject as an arch.	26
GrabSystem.Triggers.GrabArea	
A simple component that uses the 'OnTriggerEnter' and 'OnTriggerExit' callbacks to allow a ConditionalGrabber that enters a trigger to automatically grab a specified GrabbableObject when it fails to grab anything else after a grab attempt while it remains in the trigger.	28
GrabSystem.GrabbableChildObject	
A component that makes a GrabbableObject grabbable via a child collider.	29
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A component that makes an object grabbable by a Grabber	30

GrabSystem.Grabber	
The base component for all grabbers.	36
GrabSystem.Triggers.GrabberReleaseArea	
A simple component that uses the 'OnTriggerEnter' and 'OnTriggerExit' callbacks to force a grabber to release on object when it enters the relevant trigger. Optionally prevents the hand from grabbing while in the release.	44
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GrabSystem.Poses.MaintainOffsetByProjectedAngle.GrabInfo	46
GrabSystem.GrabbableObject.GrabModeChangedUnityEvent	
Arg0: GrabMode - The old grab mode. Arg1: GrabMode - The new grab mode.	46
GrabSystem.GrabUnityEvent	
Arg0: Grabber - The Grabber involved in this grab event. Arg1: GrabbableObject - The GrabbableObject that was involved in the event.	47
GrabSystem.Collisions.IgnoreCollidersWhileGrabbing	
A component intended to be attached to a GrabbableObject that subscribes to the relevant 'Grabbed' and 'Released' events to ignore collisions between some colliders and a Grabber while it is holding an object	47
GrabSystem.Poses.MaintainOffsetByGrabOffset	
A component that is attached to the same GameObject as a GrabbableObject (that is assumed to be using the 'Maintain Offset' grab mode) which overrides the maintain offset behaviour allowing more complex behaviours to be defined using the grab offset of the Grabber from the grabbable at the time it grabbed the GrabbableObject . NOTE: This component works in the grabbables local space by comparing the 'signed distance in direction' of the grabber from the grabbable at the time of grab using the comparator specified in the entry.	48
GrabSystem.Poses.MaintainOffsetByProjectedAngle	
A component that is attached to the same GameObject as a GrabbableObject (that is assumed to be using the 'Maintain Offset' grab mode) which overrides the maintain offset behaviour allowing more complex behaviours to be defined using the relative angle of the GrabbableObject from the Grabber at the time it is grabbed. NOTE: This component works by projecting the 'grabber' onto the local 'axis' in local space of the grabbable and getting the hands angle around that axis. . .	49
GrabSystem.RayGrabber	
A component that extends a Grabber by adding the ability to attempt grabs using a raycast. . .	51
GrabSystem.Collisions.DelayUnignoreCollidersOnRelease.UnignoreEntry	54

Chapter 4

Namespace Documentation

4.1 GrabSystem Namespace Reference

Classes

- class [ConditionalGrabber](#)
The base component for all grabbers that implement the '[ConditionalGrab\(\)](#)' method. Also implements repeat grab attempt routine functionality.
- class [DistanceCatchAssist](#)
A component that is generally NOT attached to the same GameObject as a [DistanceGrabber](#) that must be referenced to enable assisted catching. This is NOT attached to the same GameObject as a [DistanceGrabber](#) to avoid triggering unwanted components.
- class [DistanceGrabbable](#)
A component that can be attached to the same GameObject as a [GrabbableObject](#) that allows a [Grabber](#) to instantly grab it from a distance.
- class [DistanceGrabber](#)
A component that is attached to the same GameObject as a [Grabber](#) to enable it to grab [DistanceGrabbable](#) objects from a distance.
- class [ForceDistanceGrabbable](#)
A component that is derived from '[DistanceGrabbable](#)' that allows for force based grabbing of a [GrabbableObject](#) that also has a Rigidbody component. Modes: Instant - Teleports the object to the grabber and grabs it instantly (as a normal [DistanceGrabbable](#) would). Arch - Uses physics forces to pull the GameObject as an arch.
- class [GrabbableChildObject](#)
A component that makes a [GrabbableObject](#) grabbable via a child collider.
- class [GrabbableObject](#)
A component that makes an object grabbable by a [Grabber](#).
- class [Grabber](#)
The base component for all grabbers.
- class [GrabberUnityEvent](#)
Arg0: [Grabber](#) - The [Grabber](#) involved in this grab event.
- class [GrabUnityEvent](#)
Arg0: [Grabber](#) - The [Grabber](#) involved in this grab event. Arg1: [GrabbableObject](#) - The [GrabbableObject](#) that was involved in the event.
- class [RayGrabber](#)
A component that extends a [Grabber](#) by adding the ability to attempt grabs using a raycast.

4.2 GrabSystem.Collisions Namespace Reference

Classes

- class [DelayUnignoreCollidersOnRelease](#)

A simple component that references a [IgnoreCollidersWhileGrabbing](#) component that uses the 'OverrideUnignoreCollidersWhileGrabbing' delegate event to delay the unignore collisions for some time.

- class [IgnoreCollidersWhileGrabbing](#)

A component intended to be attached to a [GrabbableObject](#) that subscribes to the relevant 'Grabbed' and 'Released' events to ignore collisions between some colliders and a [Grabber](#) while it is holding an object.

4.3 GrabSystem.Delegates Namespace Reference

Functions

- delegate void [ActionRef< T >](#) (ref T pItem)

A simple delegate for events where an argument is passed by reference.

- delegate void [ActionRef< VALUE_ONE, T >](#) (VALUE_ONE pValueOne, ref T pItem)

A simple delegate for events where an argument is passed by reference.

- delegate void [ActionRef< VALUE_ONE, VALUE_TWO, T >](#) (VALUE_ONE pValueOne, VALUE_TWO pValueTwo, ref T pItem)

A simple delegate for events where an argument is passed by reference.

- delegate void [ActionRef< VALUE_ONE, VALUE_TWO, VALUE_THREE, T >](#) (VALUE_ONE pValueOne, VALUE_TWO pValueTwo, VALUE_THREE pValueThree, ref T pItem)

A simple delegate for events where an argument is passed by reference.

4.3.1 Function Documentation

4.3.1.1 ActionRef< T >()

```
delegate void GrabSystem.Delegates.ActionRef< T > (
    ref T pItem )
```

A simple delegate for events where an argument is passed by reference.

Template Parameters

<i>T</i>	The type of the passed reference.
----------	-----------------------------------

Parameters

<i>pItem</i>	The reference that was passed.
--------------	--------------------------------

4.3.1.2 ActionRef< VALUE_ONE, T >()

```
delegate void GrabSystem.Delegates.ActionRef< VALUE_ONE, T > (  
    VALUE_ONE pValueOne,  
    ref T pItem )
```

A simple delegate for events where an argument is passed by reference.

Template Parameters

<i>VALUE_ONE</i>	The type of the first passed value.
<i>T</i>	The type of the passed reference.

Parameters

<i>pValueOne</i>	The first value that was passed.
<i>pItem</i>	The reference that was passed.

4.3.1.3 ActionRef< VALUE_ONE, VALUE_TWO, T >()

```
delegate void GrabSystem.Delegates.ActionRef< VALUE_ONE, VALUE_TWO, T > (  
    VALUE_ONE pValueOne,  
    VALUE_TWO pValueTwo,  
    ref T pItem )
```

A simple delegate for events where an argument is passed by reference.

Template Parameters

<i>VALUE_ONE</i>	The type of the first passed value.
<i>VALUE_TWO</i>	The type of the second passed value.
<i>T</i>	The type of the passed reference.

Parameters

<i>pValueOne</i>	The first value that was passed.
<i>pValueTwo</i>	The second value that was passed.
<i>pItem</i>	The reference that was passed.

4.3.1.4 ActionRef< VALUE_ONE, VALUE_TWO, VALUE_THREE, T >()

```
delegate void GrabSystem.Delegates.ActionRef< VALUE_ONE, VALUE_TWO, VALUE_THREE, T > (  
    VALUE_ONE pValueOne,
```

```

VALUE_TWO pValueTwo,
VALUE_THREE pValueThree,
ref T pItem )

```

A simple delegate for events where an argument is passed by reference.

Template Parameters

<code>VALUE_ONE</code>	The type of the first passed value.
<code>VALUE_TWO</code>	The type of the second passed value.
<code>VALUE_THREE</code>	The type of the third passed value.
<code>T</code>	The type of the passed reference.

Parameters

<code>pValueOne</code>	The first value that was passed.
<code>pValueTwo</code>	The second value that was passed.
<code>pValueThree</code>	The third value that was passed.
<code>pItem</code>	The reference that was passed.

4.4 GrabSystem.Math Namespace Reference

Classes

- class **FloatMath**
A public static class that provides helper functions for floating-point related math operations.
- class **VectorMath**
A static class that contains vector math functions.

4.5 GrabSystem.Poses Namespace Reference

Classes

- class [MaintainOffsetByGrabOffset](#)
A component that is attached to the same GameObject as a [GrabbableObject](#) (that is assumed to be using the 'Maintain Offset' grab mode) which overrides the maintain offset behaviour allowing more complex behaviours to be defined using the grab offset of the [Grabber](#) from the grabbable at the time it grabbed the [GrabbableObject](#). NOTE: This component works in the grabbables local space by comparing the 'signed distance in direction' of the grabber from the grabbable at the time of grab using the comparator specified in the entry.
- class [MaintainOffsetByProjectedAngle](#)
A component that is attached to the same GameObject as a [GrabbableObject](#) (that is assumed to be using the 'Maintain Offset' grab mode) which overrides the maintain offset behaviour allowing more complex behaviours to be defined using the relative angle of the [GrabbableObject](#) from the [Grabber](#) at the time it is grabbed. NOTE: This component works by projecting the 'grabber' onto the local 'axis' in local space of the grabbable and getting the hands angle around that axis.

4.6 GrabSystem.Structures Namespace Reference

Classes

- class [AxisToggleSettings](#)
- struct [FloatMinMax](#)

4.7 GrabSystem.Triggers Namespace Reference

Classes

- class [GrabArea](#)
A simple component that uses the 'OnTriggerEnter' and 'OnTriggerExit' callbacks to allow a [ConditionalGrabber](#) that enters a trigger to automatically grab a specified [GrabbableObject](#) when it fails to grab anything else after a grab attempt while it remains in the trigger.
- class [GrabberReleaseArea](#)
A simple component that uses the 'OnTriggerEnter' and 'OnTriggerExit' callbacks to force a grabber to release on object when it enters the relevant trigger. Optionally prevents the hand from grabbing while in the release.

4.8 GrabSystem.Utility Namespace Reference

Classes

- class **ComponentUtility**
A public static class that provides Component related helper functions.

Chapter 5

Class Documentation

5.1 GrabSystem.Structures.AxisToggleSettings Class Reference

Public Attributes

- bool **x**
Toggle the setting for the
- bool **y**
Toggle the setting for the
- bool **z**
Toggle the setting for the

The documentation for this class was generated from the following file:

- AxisToggleSettings.cs

5.2 GrabSystem.RayGrabber.BoxCastData Class Reference

Public Member Functions

- **BoxCastData** ()
Instantiaets a [BoxCastData](#) instance.
- **BoxCastData** ([BoxCastData](#) pOther)
Instantiates a copy of some other [BoxCastData](#), pOther.

Public Attributes

- Transform **origin**
The
- float **boxHeight**
How high of a box to cast. This is the height of the box in units
- float **boxWidth**
The width of the box to cast in units.
- float **boxLength**
The length of the box to cast in units.

5.2.1 Constructor & Destructor Documentation

5.2.1.1 BoxCastData()

```
GrabSystem.RayGrabber.BoxCastData.BoxCastData (
    BoxCastData pOther )
```

Instantiates a copy of some other [BoxCastData](#), pOther.

Parameters

pOther	
------------------------	--

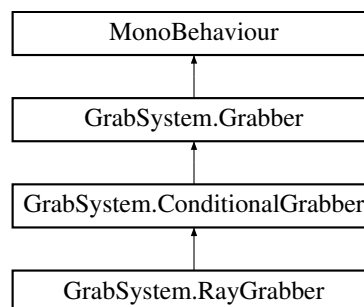
The documentation for this class was generated from the following file:

- RayGrabber.cs

5.3 GrabSystem.ConditionalGrabber Class Reference

The base component for all grabbers that implement the '[ConditionalGrab\(\)](#)' method. Also implements repeat grab attempt routine functionality.

Inheritance diagram for GrabSystem.ConditionalGrabber:



Public Member Functions

- void **StartGrabRoutine** ()
Starts a grab routine that continuously attempts grabs til successful or stopped.
- void **StopGrabRoutine** ()
Stops any ongoing continous grab attempt routine.
- void **AttemptGrab** ()
A simple wrapper for [TryGrab\(\)](#) that has no return value. Useful for use with Unity Editor events.
- [GrabbableObject TryGrab](#) ()
A simpler wrapper for [TryGrab\(PhysicsSceneUtility.GetLocalPhysicsScene\(gameObject\)\)](#).
- [GrabbableObject TryGrab](#) (PhysicsScene pPhysicsScene)

Attempts a conditional grab using the overrideable '[ConditionalGrab\(PhysicsScene\)](#)' method.

- bool [CheckInterruptGrabRoutine](#) ()
Polls all '[ShouldInterruptGrabRoutineDelegate](#)' event followers (and any other and returns true if this component will interrupt a grab based on the subscribers to the '[ShouldInterruptGrabRoutineDelegate](#)' event, otherwise false. Note that interrupted grab routines will not end the routine, they will simply prevent the routine from attempting a grab.
- abstract [GrabbableObject](#) [ConditionalGrab](#) (PhysicsScene pPhysicsScene)
Performs a conditional grab attempt in the specified PhysicsScene and returns the [GrabbableObject](#) that was grabbed, otherwise null.

Public Attributes

- [GrabberUnityEvent](#) **PreGrabAttempted**
An event that is invoked just before a grab is attempted.
- [GrabberUnityEvent](#) **TryGrabFallback**
An event that is invoked before grab success or fail event to provide a chance to implement grab fallback behaviours.
- [GrabberUnityEvent](#) **GrabAttemptRepeated**
Invoked whenever a grab routine initiates a repeat grab attempt.
- [GrabberUnityEvent](#) **GrabAttemptSucceeded**
Invoked whenever a grab attempt succeeds at grabbing a valid [GrabbableObject](#).
- [GrabberUnityEvent](#) **GrabAttemptFailed**
Invoked whenever a grab attempt fails at grabbing a valid [GrabbableObject](#).

Properties

- Coroutine **GrabRoutine** [get]
A reference to the coroutine for an ongoing continous grab attempt or null if none ongoing.

Events

- ActionRef< [Grabber](#), bool > **ShouldInterruptGrabRoutineDelegate**
An event that is invoked to provide an opportunity to interrupt a grab routine. Arg0: [Grabber](#) - the grabber whose grab routine is testing if it should be interrupted. Arg1: ref bool - a reference to the boolean value that controls whether or not the grab routine should be interrupted. Leave it unchanged for default behaviour.
- ActionRef< [Grabber](#), bool > **ShouldAllowConditionalGrabDelegate**
An event that is invoked to provide an opportunity to override whether or not the component should invoke [ConditionalGrab](#) during a [TryGrab](#). Arg0: [Grabber](#) - the grabber who is considering a conditional grab. Arg1: ref bool - a reference to the boolean value that controls whether or not to allow [ConditionalGrab](#) to be invoked by [TryGrab](#).

Additional Inherited Members

5.3.1 Detailed Description

The base component for all grabbers that implement the '[ConditionalGrab\(\)](#)' method. Also implements repeat grab attempt routine functionality.

The '[ShouldInterruptGrab](#)' event uses a reference to a boolean value as its first argument, if a subscribing callback makes this boolean true any conditional grab will be interrupted. Note that interrupting a grab routine does not end the grab routine, it simply prevents it from attempting any grab. Author: Mathew Aloisio

5.3.2 Member Function Documentation

5.3.2.1 CheckInterruptGrabRoutine()

```
bool GrabSystem.ConditionalGrabber.CheckInterruptGrabRoutine ( )
```

Polls all 'ShouldInterruptGrabRoutineDelegate' event followers (and any other and returns true if this component will interrupt a grab based on the subscribers to the 'ShouldInterruptGrabRoutineDelegate' event, otherwise false. Note that interrupted grab routines will not end the routine, they will simply prevent the routine from attempting a grab.

Returns

true if this component will interrupt a grab routine based on the subscribers to the 'ShouldInterruptGrab' event, otherwise false.

5.3.2.2 ConditionalGrab()

```
abstract GrabbableObject GrabSystem.ConditionalGrabber.ConditionalGrab (
    PhysicsScene pPhysicsScene ) [pure virtual]
```

Performs a conditional grab attempt in the specified PhysicsScene and returns the [GrabbableObject](#) that was grabbed, otherwise null.

Returns

the [GrabbableObject](#) that was grabbed, otherwise null.

Implemented in [GrabSystem.RayGrabber](#).

5.3.2.3 TryGrab() [1/2]

```
GrabbableObject GrabSystem.ConditionalGrabber.TryGrab ( )
```

A simpler wrapper for TryGrab(PhysicsSceneUtility.GetLocalPhysicsScene(gameObject)).

Returns

A reference to the [GrabbableObject](#) that was grabbed, otherwise null.

5.3.2.4 TryGrab() [2/2]

```
GrabbableObject GrabSystem.ConditionalGrabber.TryGrab (
    PhysicsScene pPhysicsScene )
```

Attempts a conditional grab using the overrideable '[ConditionalGrab\(PhysicsScene\)](#)' method.

Parameters

<i>pPhysicsScene</i>	The PhysicsScene to try the grab in.
----------------------	--------------------------------------

Returns

A reference to the [GrabbableObject](#) that was grabbed, otherwise null.

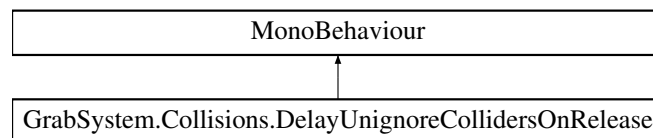
The documentation for this class was generated from the following file:

- ConditionalGrabber.cs

5.4 GrabSystem.Collisions.DelayUnignoreCollidersOnRelease Class Reference

A simple component that references a [IgnoreCollidersWhileGrabbing](#) component that uses the 'OverrideUnignore↵ Delegate' delegate event to delay the unignore collisions for some time.

Inheritance diagram for GrabSystem.Collisions.DelayUnignoreCollidersOnRelease:



Classes

- struct [UnignoreEntry](#)

Public Attributes

- float **unignoreIn** = 0.375f
Unignore collisions this many seconds after
- [IgnoreCollidersWhileGrabbing](#) **ignoreComponent**
A reference to the [IgnoreCollidersWhileGrabbing](#) component associated with this component.

5.4.1 Detailed Description

A simple component that references a [IgnoreCollidersWhileGrabbing](#) component that uses the 'OverrideUnignore↵ Delegate' delegate event to delay the unignore collisions for some time.

This component references an [IgnoreCollidersWhileGrabbing](#) component instead of being forced onto the same GameObject because frequently a developer may want to disable the relevant GameObject after release, this allows this component to be put on a separate non-deactivated GameObject while still allowing the unignore collisions delay. Author: Mathew Aloisio

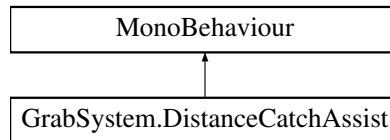
The documentation for this class was generated from the following file:

- DelayUnignoreCollidersOnRelease.cs

5.5 GrabSystem.DistanceCatchAssist Class Reference

A component that is generally NOT attached to the same GameObject as a [DistanceGrabber](#) that must be referenced to enable assisted catching. This is NOT attached to the same GameObject as a [DistanceGrabber](#) to avoid triggering unwanted components.

Inheritance diagram for GrabSystem.DistanceCatchAssist:



Public Attributes

- [DistanceGrabber](#) **distanceGrabber**
The distance grabber this component assists with catching.
- bool **snapGrab**
Should this component snap grabbable objects that is assisted catches to the grab target using the distance grab hit point?
- [GrabUnityEvent](#) **CatchAssisted**
An event that is invoked when a catch assist is compelled.
lnArg

5.5.1 Detailed Description

A component that is generally NOT attached to the same GameObject as a [DistanceGrabber](#) that must be referenced to enable assisted catching. This is NOT attached to the same GameObject as a [DistanceGrabber](#) to avoid triggering unwanted components.

When a distance grabbable that is being distance grabbed by the relevant distance grabber triggers this component an instantenous grab is performed. Author: Mathew Aloisio

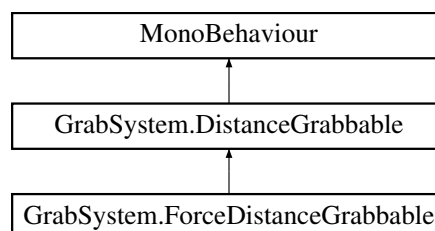
The documentation for this class was generated from the following file:

- DistanceCatchAssist.cs

5.6 GrabSystem.DistanceGrabbable Class Reference

A component that can be attached to the same GameObject as a [GrabbableObject](#) that allows a [Grabber](#) to instantly grab it from a distance.

Inheritance diagram for GrabSystem.DistanceGrabbable:



Public Member Functions

- bool [CanDistanceGrab](#) ([Grabber](#) pGrabber)

Returns true if pGrabber can distance grab this, otherwise false. NOTE: This does not check all conditions for the [Grabber](#), the [Grabber](#) may have its own conditions not specific to this Grabbable. See '[DistanceGrabber.CanDistanceGrab\(DistanceGrabbable\)](#)'.
- void [InstantGrab](#) (Vector3 pHitPoint)

Instantly grabs a distance grabbable.
- void [SetTarget](#) ([DistanceGrabber](#) pGrabber, Vector3 pHitPoint)

Targets the specified [DistanceGrabber](#), or none if null. The distance grabbable will immediate start being 'grabbed' (teleported or pulled).
- void **ClearTarget** ()

Clears any target of the [DistanceGrabbable](#).
- void [Highlight](#) ([Grabber](#) pGrabber)

Tells this [DistanceGrabbable](#) that pGrabber has started highlighting it (if it is not already highlighting it).
- void [Unhighlight](#) ([Grabber](#) pGrabber)

Tells this [DistanceGrabbable](#) that pGrabber has stopped highlighting it (if it is already highlighting it).
- [Grabber](#) [GetHighlightedBy](#) (int pIndex)

Returns the [Grabber](#) that is highlighting this distance grabbable by index.

Public Attributes

- bool **rotate**

Should this distance grabbable rotate towards the specified rotateTo?
- Vector3 **rotateTo**

The local space euler angles relative to the [DistanceGrabber.GrabTarget](#) doing the distance grabbing the object should rotate towards while being distance grabbed.
- [GrabUnityEvent](#) **DistanceGrabStarted**

An event that is invoked whenever a distance grab is started.
lnArg
- [GrabUnityEvent](#) **DistanceGrabStopped**

An event that is invoked whenever a distance grab is stopped.
lnArg
- [GrabUnityEvent](#) **InstantGrabbed**

An event that is invoked whenever a distance grabbable is
- [GrabberUnityEvent](#) **StartedHighlighting**

An event that is invoked when a [Grabber](#) starts highlighting this distance grabbable.
lnArg
- [GrabberUnityEvent](#) **StoppedHighlighting**

An event that is invoked when a [Grabber](#) stops highlighting this distance grabbable.
lnArg
- [GrabberUnityEvent](#) **AllStoppedHighlighting**

An event that is invoked when the last [Grabber](#) stops highlighting this distance grabbable.
lnArg

Protected Member Functions

- virtual void **Awake** ()
- virtual void **OnEnable** ()
- virtual void **OnDisable** ()
- virtual void [OnDistanceGrabStarted](#) (Vector3 pHitPoint)

Invoked when a distance grab is started.
- virtual void **OnDistanceGrabStopped** ()

Invoked when a distance grab is stopped.

Properties

- [GrabbableObject](#) **Grabbable** [get]
The [GrabbableObject](#) component that this component is converting into a distance grabbable.
- [DistanceGrabber](#) **Target** [get]
A reference to the [GrabbableObject](#) being targetted by distance [DistanceGrabbable](#), otherwise null.
- [DistanceGrabber](#) **LastTarget** [get]
The last set target of the distance grabbable, otherwise null if there was none. This reference remains valid after the distance grab is stopped. (This updates immediately when a new valid target is set and will at that point be equal to Target.)
- [Vector3](#) **TargetLocalHitPoint** [get]
The point the target hit this distance grabbable at during selection in the distance grabbables local space. This is only valid while Target != null.
- [int](#) **HighlightedByCount** [get]
Returns the number of Grabbers currently highlighting this distance grabbable.

Events

- [ActionRef](#)< [Grabber](#), [GrabbableObject](#), bool > **CanDistanceGrabDelegate**
A delegate even that allows distance grab permissions to be overridden by subscribing to this event and modifying the 'ref bool' argument (argument 3). Arg0: [Grabber](#) - The [Grabber](#) doing the grabbing. This [Grabber](#) is guarenteed to have a [DistanceGrabber](#) component. Arg1: [GrabbableObject](#) - The [GrabbableObject](#) being distance grabbed. This [GrabbableObject](#) is guarenteed to have a [DistanceGrabbable](#) component. Arg2: ref bool - The reference to a boolean that when changed to false (or true) forces distance grabs for this object o be denied (or approved).

5.6.1 Detailed Description

A component that can be attached to the same GameObject as a [GrabbableObject](#) that allows a [Grabber](#) to instantly grab it from a distance.

Author: Mathew Aloisio

5.6.2 Member Function Documentation

5.6.2.1 CanDistanceGrab()

```
bool GrabSystem.DistanceGrabbable.CanDistanceGrab (
    Grabber pGrabber )
```

Returns true if pGrabber can distance grab this, otherwise false. NOTE: This does not check all conditions for the [Grabber](#), the [Grabber](#) may have its own conditions not specific to this Grabbable. See '[DistanceGrabber.CanDistanceGrab\(DistanceGrabbable\)](#)'.

By default this method uses 'pGrabber.CanGrab(Grabbable)' to check distance grab permission. This method invokes the callback 'CanDistanceGrabDelegate' which gives scripts an opportunity to override distance grab permissions.

Parameters

<i>pGrabber</i>	
-----------------	--

Returns

true if pGrabber can distance grab this distance grabbable, otherwise false.

5.6.2.2 GetHighlightedBy()

```
Grabber GrabSystem.DistanceGrabbable.GetHighlightedBy (
    int pIndex )
```

Returns the [Grabber](#) that is highlighting this distance grabbable by index.

Parameters

<i>pIndex</i>	
---------------	--

5.6.2.3 Highlight()

```
void GrabSystem.DistanceGrabbable.Highlight (
    Grabber pGrabber )
```

Tells this [DistanceGrabbable](#) that pGrabber has started highlighting it (if it is not already highlighting it).

Parameters

<i>pGrabber</i>	
-----------------	--

5.6.2.4 InstantGrab()

```
void GrabSystem.DistanceGrabbable.InstantGrab (
    Vector3 pHitPoint )
```

Instantly grabs a distance grabbable.

Parameters

<i>pHitPoint</i>	The 'hit point' the grabber targeted.
------------------	---------------------------------------

5.6.2.5 OnDistanceGrabStarted()

```
virtual void GrabSystem.DistanceGrabbable.OnDistanceGrabStarted (
    Vector3 pHitPoint ) [protected], [virtual]
```

Invoked when a distance grab is started.

Parameters

<i>pHitPoint</i>	The world space 'hit point' the grabber targeted in world space.
------------------	--

Reimplemented in [GrabSystem.ForceDistanceGrabbable](#).

5.6.2.6 SetTarget()

```
void GrabSystem.DistanceGrabbable.SetTarget (
    DistanceGrabber pGrabber,
    Vector3 pHitPoint )
```

Targets the specified [DistanceGrabber](#), or none if null. The distance grabbable will immediate start being 'grabbed' (teleported or pulled).

Parameters

<i>pGrabber</i>	The DistanceGrabber that is pulling this distance grabbable.
<i>pHitPoint</i>	The 'hit point' the grabber targeted.

5.6.2.7 Unhighlight()

```
void GrabSystem.DistanceGrabbable.Unhighlight (
    Grabber pGrabber )
```

Tells this [DistanceGrabbable](#) that pGrabber has stopped highlighting it (if it is already highlighting it).

Parameters

<i>pGrabber</i>	
-----------------	--

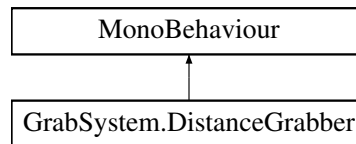
The documentation for this class was generated from the following file:

- DistanceGrabbable.cs

5.7 GrabSystem.DistanceGrabber Class Reference

A component that is attached to the same GameObject as a [Grabber](#) to enable it to grab [DistanceGrabbable](#) objects from a distance.

Inheritance diagram for GrabSystem.DistanceGrabber:



Public Member Functions

- bool [CanDistanceGrab](#) ([DistanceGrabbable](#) pGrabbable)
Returns true if this [Grabber](#) can distance grab pGrabbable, otherwise false.
- void [StartPull](#) ([DistanceGrabbable](#) pGrabbable)
Stops any existing pull operation and starts pulling the [DistanceGrabbable](#), pGrabbable. NOTE: This method does not check grab permissions, see 'bool CanGrab(DistanceGrabbable)'.
- void [StartPull](#) ([DistanceGrabbable](#) pGrabbable, Vector3 pHitPoint)
Stops any existing pull operation and starts pulling the [DistanceGrabbable](#), pGrabbable. NOTE: This method does not check grab permissions, see 'bool CanGrab(DistanceGrabbable)'.
- void **StopPull** ()
Stops any existing pull operation.

Public Attributes

- [GrabUnityEvent](#) **StartedPull**
An event that is invoked when this component starts pulling a distance grabbable.
InArg
- [GrabUnityEvent](#) **StoppedPull**
An event that is invoked when this component stops pulling a distance grabbable.
InArg
- [GrabUnityEvent](#) **InstantPulled**
An event that is invoked if a distance grabbable is instant pulled to the grabber.
InArg

Properties

- [Grabber](#) **Grabber** [get]
A reference to the [Grabber](#) associated with this [DistanceGrabber](#).
- Transform **GrabTarget** [get]
A reference to the Transform this distance grabber will teleport or pull distance grabbables to.
- [DistanceGrabbable](#) **Pulling** [get]
A reference to the [DistanceGrabbable](#) currently being pulled by this component, or null.

Events

- ActionRef< [Grabber](#), [GrabbableObject](#), bool > **CanDistanceGrabDelegate**

A delegate even that allows distance grab permissions to be overridden by subscribing to this event and modifying the 'ref bool' argument (argument 3). Arg0: [Grabber](#) - The [Grabber](#) doing the grabbing. This [Grabber](#) is guarenteed to have a [DistanceGrabber](#) component. Arg1: [GrabbableObject](#) - The [GrabbableObject](#) being distance grabbed. This [GrabbableObject](#) is guarenteed to have a [DistanceGrabbable](#) component. Arg2: ref bool - The reference to a boolean that when changed to false (or true) forces distance grabs for this object o be denied (or approved).

5.7.1 Detailed Description

A component that is attached to the same GameObject as a [Grabber](#) to enable it to grab [DistanceGrabbable](#) objects from a distance.

Author: Mathew Aloisio

5.7.2 Member Function Documentation

5.7.2.1 CanDistanceGrab()

```
bool GrabSystem.DistanceGrabber.CanDistanceGrab (
    DistanceGrabbable pGrabbable )
```

Returns true if this [Grabber](#) can distance grab pGrabbable, otherwise false.

Parameters

pGrabbable	
----------------------------	--

Returns

true if this [Grabber](#) can distance grab pGrabbable, otherwise false.

5.7.2.2 StartPull() [1/2]

```
void GrabSystem.DistanceGrabber.StartPull (
    DistanceGrabbable pGrabbable )
```

Stops any existing pull operation and starts pulling the [DistanceGrabbable](#), pGrabbable. NOTE: This method does not check grab permissions, see 'bool CanGrab(DistanceGrabbable)'.

Parameters

pGrabbable	
----------------------------	--

5.7.2.3 StartPull() [2/2]

```
void GrabSystem.DistanceGrabber.StartPull (
    DistanceGrabbable pGrabbable,
    Vector3 pHitPoint )
```

Stops any existing pull operation and starts pulling the [DistanceGrabbable](#), pGrabbable. NOTE: This method does not check grab permissions, see 'bool CanGrab(DistanceGrabbable)'.

Parameters

<i>pGrabbable</i>	
<i>pHitPoint</i>	The hit point the pull was started from.

The documentation for this class was generated from the following file:

- DistanceGrabber.cs

5.8 GrabSystem.Poses.MaintainOffsetByGrabOffset.Entry Class Reference

Public Attributes

- Transform **maintainOffsetTransform**
The Transform to maintain offset using when this entrys conditions are met first.
- Vector3 **localOffset** = Vector3.zero
The local space
- Vector3 **direction**
The direction to test the grabber distance in
- FloatMath.Comparator **comparator**
The comparator to use when testing distance
- float **distanceThreshold**
The minimum distance in the specified direction the grabber must be to satisfy the conditions.
- [GrabberUnityEvent](#) **EntryUsed**
An event that is invoked when this entry is used by a grabber.
lnArg

The documentation for this class was generated from the following file:

- MaintainOffsetByGrabOffset.cs

5.9 GrabSystem.Poses.MaintainOffsetByProjectedAngle.Entry Class Reference

Public Attributes

- Transform **maintainOffsetTransform**
The Transform to maintain offset using when this entrys conditions are met first.
- Vector3 **upAxis** = Vector3.up
The upwards direction vector to test against.
- Vector3 **angleAxis** = Vector3.right
The direction vector axis that defines the plane to project the grabber onto.
- Transform **centerPoint**
The center point Transform to get the angle around. [If null
- [FloatMinMax](#) **angleRange**
The angle range the
- [GrabberUnityEvent](#) **EntryUsed**
An event that is invoked when this entry is used by a grabber.
lnArg

The documentation for this class was generated from the following file:

- MaintainOffsetByProjectedAngle.cs

5.10 GrabSystem.Structures.FloatMinMax Struct Reference

Public Attributes

- float **minimum**
The minimum value.
- float **maximum**
The maximum value.

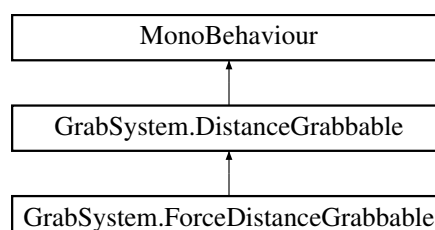
The documentation for this struct was generated from the following file:

- FloatMinMax.cs

5.11 GrabSystem.ForceDistanceGrabbable Class Reference

A component that is derived from '[DistanceGrabbable](#)' that allows for force based grabbing of a [GrabbableObject](#) that also has a Rigidbody component. Modes: Instant - Teleports the object to the grabber and grabs it instantly (as a normal [DistanceGrabbable](#) would). Arch - Uses physics forces to pull the GameObject as an arch.

Inheritance diagram for GrabSystem.ForceDistanceGrabbable:



Public Types

- enum **PullMode**

Public Member Functions

- Vector3 **CalculateTrajectory** (Vector3 pOrigin, Vector3 pTarget, float pArch)
Calculates the trajectory velocity from the target to the origin and returns it as a Vector3.

Public Attributes

- PullMode **pullMode** = PullMode.Arch
*The pull mode to use for the distance grabbable.
Instant*
- bool **stopPullOnCollision** = true
Should this distance grabbable stop being pulled after a collision collisionGrace seconds after pulling starts?
- float **collisionEnterGrace** = 0.2f
The number of seconds of collision enter grace granted before de
- float **collisionStayGrace** = 0.8f
The number of seconds of collision stay grace granted before de
- float **archMultiplier** = 0.7f
The arch strength multiplication factor for the distance grabbable.
- float **pullSpeed** = 1f
The pull speed scalar for the distance grabbable.
- float **pullVelocityScalar** = 10f
A scalar that can be used to framerate independently scale pull velocity.
- float **rotateSpeed** = 540f
If

Protected Member Functions

- virtual void **FixedUpdate** ()
- virtual void **OnCollisionEnter** (Collision pCollision)
- virtual void **OnCollisionStay** (Collision pCollision)
- override void **OnDistanceGrabStarted** (Vector3 pHitPoint)
Invoked when a distance grab is started.

Additional Inherited Members

5.11.1 Detailed Description

A component that is derived from 'DistanceGrabbable' that allows for force based grabbing of a [GrabbableObject](#) that also has a Rigidbody component. Modes: Instant - Teleports the object to the grabber and grabs it instantly (as a normal [DistanceGrabbable](#) would). Arch - Uses physics forces to pull the GameObject as an arch.

Author: Mathew Aloisio

5.11.2 Member Function Documentation

5.11.2.1 CalculateTrajectory()

```
Vector3 GrabSystem.ForceDistanceGrabbable.CalculateTrajectory (
    Vector3 pOrigin,
    Vector3 pTarget,
    float pArch )
```

Calculates the trajectory velocity from the target to the origin and returns it as a Vector3.

Parameters

<i>pOrigin</i>	
<i>pTarget</i>	
<i>pArch</i>	

Returns

a Vector3 representing the trajectory velocity from the target to the origin.

5.11.2.2 OnDistanceGrabStarted()

```
override void GrabSystem.ForceDistanceGrabbable.OnDistanceGrabStarted (
    Vector3 pHitPoint ) [protected], [virtual]
```

Invoked when a distance grab is started.

Parameters

<i>pHitPoint</i>	The world space 'hit point' the grabber targeted in world space.
------------------	--

Reimplemented from [GrabSystem.DistanceGrabbable](#).

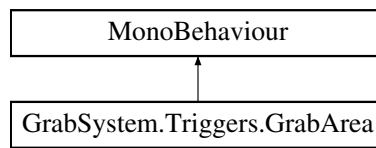
The documentation for this class was generated from the following file:

- ForceDistanceGrabbable.cs

5.12 GrabSystem.Triggers.GrabArea Class Reference

A simple component that uses the 'OnTriggerEnter' and 'OnTriggerExit' callbacks to allow a [ConditionalGrabber](#) that enters a trigger to automatically grab a specified [GrabbableObject](#) when it fails to grab anything else after a grab attempt while it remains in the trigger.

Inheritance diagram for GrabSystem.Triggers.GrabArea:



Public Attributes

- [GrabbableObject](#) **grabbable**
A reference to the [GrabbableObject](#) that can be grabbed using this [GrabArea](#).
- bool **checkRigidbody** = true
If not found on the triggering Collider

5.12.1 Detailed Description

A simple component that uses the 'OnTriggerEnter' and 'OnTriggerExit' callbacks to allow a [ConditionalGrabber](#) that enters a trigger to automatically grab a specified [GrabbableObject](#) when it fails to grab anything else after a grab attempt while it remains in the trigger.

Author: Mathew Aloisio

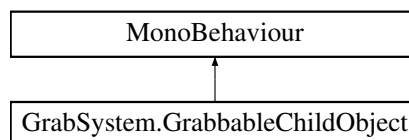
The documentation for this class was generated from the following file:

- GrabArea.cs

5.13 GrabSystem.GrabbableChildObject Class Reference

A component that makes a [GrabbableObject](#) grabbable via a child collider.

Inheritance diagram for GrabSystem.GrabbableChildObject:



Public Attributes

- [GrabbableObject](#) **grabbable**
A reference to the [GrabbableObject](#) that this component handles grab events for.

5.13.1 Detailed Description

A component that makes a [GrabbableObject](#) grabbable via a child collider.

Author: Mathew Aloisio

The documentation for this class was generated from the following file:

- GrabbableChildObject.cs

5.14 GrabSystem.RayGrabber.GrabbableEntry Struct Reference

Public Attributes

- [GrabbableObject](#) **grabbable**
The [GrabbableObject](#) reference for the pair.
- Collider **collider**
The Collider that may be grabbed.
- float **distance**
The distance of the grabbable according to the pair.

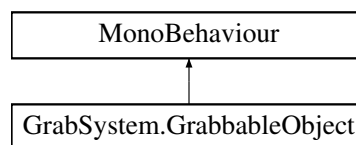
The documentation for this struct was generated from the following file:

- RayGrabber.cs

5.15 GrabSystem.GrabbableObject Class Reference

A component that makes an object grabbable by a [Grabber](#).

Inheritance diagram for GrabSystem.GrabbableObject:



Classes

- class [GrabModeChangedUnityEvent](#)
Arg0: GrabMode - The old grab mode. Arg1: GrabMode - The new grab mode.

Public Types

- enum [GrabMode](#)
The mode used when grabbing this object.

Public Member Functions

- void **SetPositionAndRotation** (Vector3 pPosition, Quaternion pRotation)
Sets the position and rotation of this [GrabbableObject](#) (and its [Rigidbody\(s\)](#) if relevant.)
- void **UpdateCachedMass** ()
If this grabbable has a valid Rigidbody overrides the cached 'default' mass value for the object with 'Rigidbody.mass'.
- void **ForceGrabbersRelease** ()
Forces all Grabbers to release this [GrabbableObject](#).
- void **ForceGrabbersReleaseNoThrow** ()
Forces all Grabbers to release this [GrabbableObject](#) without throwing.
- Vector3 **GetGrabOffset** (Grabber pGrabber)
Returns a Vector3 representing the grab offset in the pGrabbers local space for this [GrabbableObject](#). Takes into account axis flip settings.
- Vector3 **GetGrabEulerOffset** (Grabber pGrabber)
Returns a Vector3 representing the grab euler offset in the pGrabbers local space for this [GrabbableObject](#). Takes into account axis flip settings.
- Grabber **GetHeldBy** (int pIndex)
Returns the [Grabber](#) in the given 'held by' index.
- void **MaintainOffsetFrom** (Grabber pGrabber, Transform pMaintainTransform)
Maintains a pGrabber at pMaintainTransform.
- void **SetGrabEnabled** (bool pEnabled)
Sets the 'grabEnabled' field for the grabbable object. Useful for use with Unity Editor events.
- void **SetGrabMode** (GrabMode pGrabMode)
- void **UseMoveObjectGrabMode** ()
Forces the [GrabbableObject](#) to use the 'MoveObject' grab mode.
- void **UseMaintainOffsetGrabMode** ()
Forces the [GrabbableObject](#) to use the 'MaintainOffset' grab mode.
- virtual bool **CanGrab** (Grabber pGrabber)
Returns true if this [GrabbableObject](#) can be grabbed by pGrabber, otherwise false. (NOTE: This only checks conditions relevant to the grabbable object, the [Grabber](#) may have its own conditions.)

Public Attributes

- bool **grabEnabled** = true
Can this grabbable object be grabbed?
- bool **makeKinematic**
Should this Grabbable always be kinematic when grabbed?
- bool **makeWeightless**
Should this Grabbable be made weightless when grabbed?
- int **grabPriority** = 0
The grab priority for this grabbable object. When multiple grabbables meet the grab conditions for a [Grabber](#) priority will be considered first
- [GrabMode](#) **grabMode** = GrabMode.MaintainOffset
*The mode used when this object is grabbed.
lnNone*
- bool **singleGrabberGrab**
Should this object only be grabbable by a single [Grabber](#) at a time?
- bool **grabSwap**
- float **grabBreakForce** = 5000f
The force required to break the grab joint.
- float **grabBreakTorque** = 5000f

- The torque required to break the grab joint.*

 - ConfigurableJoint **grabJointPrefab**
 - bool **keepGrabOffset**
 - When in*
 - Vector3 **grabOffset**
 - When in*
 - Vector3 **grabEulerOffset**
 - When in*
 - float **throwMultiplier** = 1f
 - The multiplier for throw force being applied to this object.*
 - [GrabUnityEvent](#) **PreGrabbed**
 - An event that is invoked just before this grabbable object is grabbed.*
 - lnArg*
 - [GrabUnityEvent](#) **Grabbed**
 - An event that is invoked when this grabbable object is grabbed.*
 - lnArg*
 - [GrabUnityEvent](#) **PreReleased**
 - An event that is invoked just before this grabbable object is released.*
 - lnArg*
 - [GrabUnityEvent](#) **Released**
 - An event that is invoked when this grabbable object is released.*
 - lnArg*
 - [GrabUnityEvent](#) **LastGrabberReleased**
 - An event that is invoked when the last [Grabber](#) that is holding this grabbable object releases it.*
 - lnArg*
 - [GrabUnityEvent](#) **GrabJointBroken**
 - An event that is invoked after a*
 - [GrabModeChangedUnityEvent](#) **GrabModeChanged**
 - An event that is invoked every time the grabMode of this [GrabbableObject](#) is changed using the*
 - ActionRef< [Grabber](#), [GrabbableObject](#), bool > **MaintainOffsetOverrideDelegate**
 - A delegate that allows the 'Maintain Offset' behaviour to be overridden for this [GrabbableObject](#), when the reference-passed boolean is set to true the default 'Maintain Offset' behaviour will be skipped. This is useful for implementing things such as advanced maintained grab positions that go beyond the scope of the default settings of the [GrabbableObject](#) component. Arg0: [Grabber](#) - The [Grabber](#) who is grabbing the object. Arg1: [GrabbableObject](#) - The [GrabbableObject](#) being grabbed. Arg2: ref bool - The boolean that controls whether or not to skip the default 'Maintain Offset' behaviour. When true the default behaviour is skipped.*
 - ActionRef< [Grabber](#), [GrabbableObject](#), Vector3 > **OverrideGrabOffsetDelegate**
 - A delegate that is invoked whenever [GetGrabOffset\(Grabber\)](#) is invoked. This is useful for implementing custom behaviours for overriding a grab offset for a [GrabbableObject](#) via code. Arg0: [Grabber](#) - The [Grabber](#) who is grabbing the object. Arg1: [GrabbableObject](#) - The [GrabbableObject](#) being grabbed. Arg2: ref Vector3 - A reference to the returned grab offset in the grabbers local space.*
 - ActionRef< [Grabber](#), [GrabbableObject](#), Vector3 > **OverrideGrabEulerOffsetDelegate**
 - A delegate that is invoked whenever [GetGrabEulerOffset\(Grabber\)](#) is invoked. This is useful for implementing custom behaviours for overriding a grab euler angle offset for a [GrabbableObject](#) via code. Arg0: [Grabber](#) - The [Grabber](#) who is grabbing the object. Arg1: [GrabbableObject](#) - The [GrabbableObject](#) being grabbed. Arg2: ref Vector3 - A reference to the returned grab euler offset in the grabbers local space.*
 - ActionRef< [Grabber](#), [GrabbableObject](#), bool > **CanGrabDelegate**
 - A delegate that is invoked whenever '[CanGrab\(Grabber\)](#)' is invoked. This is useful for overriding grabbability for specific [GrabbableObject](#)s by subscribing to this event and modifying the 'ref bool' value. The value may be left unmodified to pass-through the default behaviour. Arg0: [Grabber](#) - The [Grabber](#) who is attempting to grab the object. Arg1: [GrabbableObject](#) - The [GrabbableObject](#) being grabbed. Arg2: ref bool - A reference to the boolean that decides whether or not this grabber can be grabbed by the specified grabber (at least as far as the grabbable itself is concerned).*

Protected Member Functions

- virtual void **Awake** ()
- virtual void **Start** ()
- virtual void **Update** ()

Properties

- Rigidbody **Rigidbody** [get]
The Rigidbody component that on the same GameObject as this [GrabbableObject](#), otherwise null if the grabbable is not a Rigidbody.
- int **HeldByCount** [get]
the number of hands currently holding the [GrabbableObject](#).
- Transform **MaintainGrabTransform** [get]
A reference to the Transform a hand that grabs this object will maintain it's offset relative to when enabled. If null the grab point is used instead.
- bool **CanBeGrabbed** [get]
Returns true if this [GrabbableObject](#) can be grabbed, otherwise false. (NOTE: Use [CanGrab\(Grabber\)](#) methods instead to test for grabbability by specific Grabbers, this only tests for global 'can grab' properties that apply to any Grabbers.)
- Vector3 **OffsetAtGrab** [get]
Returns 'transform.position - grabber.transform.position' at the time of grab. This offset is in the grabbers local space.
- Quaternion **RotationOffsetAtGrab** [get]
Returns the rotation offset at the time of grab. This offset is in the grabbers local space.

5.15.1 Detailed Description

A component that makes an object grabbable by a [Grabber](#).

NOTE: While in GrabMode.MoveObject the object will move based on the most recent grabbing hand. Author: Mathew Aloisio

5.15.2 Member Function Documentation

5.15.2.1 CanGrab()

```
virtual bool GrabSystem.GrabbableObject.CanGrab (
    Grabber pGrabber ) [virtual]
```

Returns true if this [GrabbableObject](#) can be grabbed by pGrabber, otherwise false. (NOTE: This only checks conditions relevant to the grabbable object, the [Grabber](#) may have its own conditions.)

Parameters

<i>pGrabber</i>	
-----------------	--

Returns

true if this [GrabbableObject](#) can be grabbed by pGrabber, otherwise false.

5.15.2.2 GetGrabEulerOffset()

```
Vector3 GrabSystem.GrabbableObject.GetGrabEulerOffset (
    Grabber pGrabber )
```

Returns a Vector3 representing the grab euler offset in the pGrabbers local space for this [GrabbableObject](#). Takes into account axis flip settings.

NOTE: This method invokes '[OverrideGrabEulerOffsetDelegate\(Grabber, GrabbableObject, ref Vector3\)](#)' for easy overriding.

Parameters

<i>pGrabber</i>	
-----------------	--

Returns

a Vector3 representing the grab euler offset in the pGrabbers local space for this [GrabbableObject](#).

5.15.2.3 GetGrabOffset()

```
Vector3 GrabSystem.GrabbableObject.GetGrabOffset (
    Grabber pGrabber )
```

Returns a Vector3 representing the grab offset in the pGrabbers local space for this [GrabbableObject](#). Takes into account axis flip settings.

NOTE: This method invokes '[OverrideGrabOffsetDelegate\(Grabber, GrabbableObject, ref Vector3\)](#)' for easy overriding.

Parameters

<i>pGrabber</i>	
-----------------	--

Returns

a Vector3 representing the grab offset in the pGrabbers local space for this [GrabbableObject](#).

5.15.2.4 GetHeldBy()

```
Grabber GrabSystem.GrabbableObject.GetHeldBy (
    int pIndex )
```

Returns the [Grabber](#) in the given 'held by' index.

Parameters

<i>pIndex</i>	
---------------	--

Returns

the [Grabber](#) in the given 'held by' index.

5.15.2.5 MaintainOffsetFrom()

```
void GrabSystem.GrabbableObject.MaintainOffsetFrom (
    Grabber pGrabber,
    Transform pMaintainTransform )
```

Maintains a *pGrabber* at *pMaintainTransform*.

Parameters

<i>pGrabber</i>	
<i>pMaintainTransform</i>	

5.15.2.6 SetGrabEnabled()

```
void GrabSystem.GrabbableObject.SetGrabEnabled (
    bool pEnabled )
```

Sets the 'grabEnabled' field for the grabbable object. Useful for use with Unity Editor events.

Parameters

<i>pEnabled</i>	
-----------------	--

5.15.2.7 SetPositionAndRotation()

```
void GrabSystem.GrabbableObject.SetPositionAndRotation (
```

```
Vector3 pPosition,
Quaternion pRotation )
```

Sets the position and rotation of this [GrabbableObject](#) (and its [Rigidbody\(s\)](#) if relevant.)

Parameters

<i>pPosition</i>	
<i>pRotation</i>	

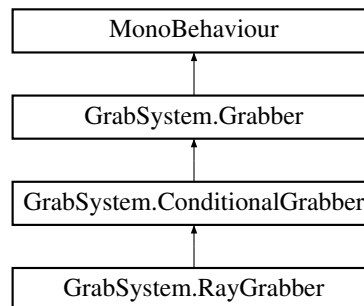
The documentation for this class was generated from the following file:

- GrabbableObject.cs

5.16 GrabSystem.Grabber Class Reference

The base component for all grabbers.

Inheritance diagram for GrabSystem.Grabber:



Public Member Functions

- void [Grab](#) ([GrabbableObject](#) pGrabbable)
Forces a [GrabbableObject](#) to be grabbed, if an object is already grabbed it is (attempted to be) released. Same as calling 'Grab(pGrabbable, null)'.
- void [Grab](#) ([GrabbableObject](#) pGrabbable, Collider pGrabCollider)
Forces a [GrabbableObject](#) to be grabbed, if an object is already being grabbed it is (attempted to be) released.
- void [GrabAtOffset](#) ([GrabbableObject](#) pGrabbable, Vector3 pOffset, Quaternion pRotation)
If pGrabbable is a movable object, teleports it to pOffset and pRotation local space (angle) offsets relative to the [Grabber's](#) 'transform', then grabs.
- Vector3 [GetTransformOffset](#) (Transform pTransform)
Returns the position of pTransform in the local space of the [Grabber's](#) transform.
- Quaternion [GetTransformRotationOffset](#) (Transform pTransform)
Returns the rotation offset of pTransform relative to the [Grabber's](#) transform.
- Vector3 [GetGrabbableOffset](#) ([GrabbableObject](#) pGrabbable)
Returns the position of pGrabbable's transform in the local space of the [Grabber's](#) transform.
- Quaternion [GetGrabbableRotationOffset](#) ([GrabbableObject](#) pGrabbable)
Returns the rotation offset of pGrabbable's transform relative to the [Grabber's](#) transform.
- void **ReleaseNoThrow** ()

- Releases the currently grabbed object without throwing it only if 'CanRelease()' returns true.*

 - void **Release** ()

Releases the currently grabbed object applying any relevant throw forces only if 'CanRelease()' returns true.
- void **ForceReleaseNoThrow** ()

Releases the currently grabbed object without throwing it.
- void **ForceRelease** ()

Releases the currently grabbed object applying any relevant throw forces.
- Vector3 **GetOffsetFromGrabber** (GrabbableObject pGrabbable)

Returns the offset of pGrabbable from 'this' Grabber as a Vector3 in local space of 'this' Grabber.
- Quaternion **GetRelativeRotationFromGrabber** (GrabbableObject pGrabbable)

Returns the relative rotation of pGrabbable from 'this' Grabber as a Quaternion.
- void **ThrowRigidbody** (Rigidbody pRigidbody, float pThrowForceMultiplier)

Calculates and applies throw forces to the specified Rigidbody.
- Joint **CreateJoint** (GrabbableObject pGrabbable, Rigidbody pRigidbody, float pBreakForce, float pBreakTorque, ConfigurableJoint pGrabJointPrefab)

Creates a physics joint between a Rigidbody and the Grabber.
- Joint **GetJointAtIndex** (int pIndex)

Returns the Joint at the given index of the joints array for this Grabber.
- void **DestroyJoint** (Joint pJoint)

Destroys a Joint and removes it from this Grabbers tracked joint list if it is in it.
- void **DestroyAllJoints** ()

Destroys all Joints that were created by this Grabber.
- void **SetGrabLocked** (bool pLocked)

Sets the 'grabLocked' field of this component. Useful for use with Unity editor events.
- virtual bool **CanGrab** (GrabbableObject pGrabbable)

Returns true if this Grabber can grab pGrabbable, otherwise false. NOTE: By default (when not overridden) it is guaranteed that this method takes into account 'CanGrabDelegate' event overrides.

Public Attributes

- bool **grabLocked**
- Lock this grabber when enabled preventing it from performing non*
- Transform **overrideAnchor**
- Allows the grabbed object anchor Transform to be overridden with a specific Transform.*
- AxisToggleSettings **flipGrabOffset**
- Allows grab offsets to be flipped per axis.*
- AxisToggleSettings **flipGrabAngle**
- Allows grab angles to be flipped per axis.*
- AxisToggleSettings **flipMaintainAngle**
- Allows maintain offset angles to be flipped per axis.*
- Rigidbody **body**
- float **throwStrength** = 1f
- The force objects at thrown with.*
- float **throwThreshold** = 0.01f
- The minimum velocity the grabber must have to apply throw forces to a Rigidbody.*
- TransformVelocityTracker **velocityTracker**
- GrabUnityEvent **PreGrabbed**
- An event that is invoked just before a Grabbable object is grabbed by this grabber.*

lnArg
- GrabUnityEvent **Grabbed**

An event that is invoked when a Grabbable object is grabbed by this grabber.
 InArg

- [GrabUnityEvent](#) **PreReleased**

An event that is invoked just before a Grabbable object is released by this grabber.
 InArg

- [GrabUnityEvent](#) **Released**

An event that is invoked when a Grabbable object is released by this grabber.
 InArg

- [GrabUnityEvent](#) **GrabJointBroken**

An event that is invoked after the

Protected Member Functions

- virtual void **Awake** ()
- virtual void **Reset** ()
- virtual void **LateUpdate** ()
- virtual void **OnDisable** ()
- virtual void **OnDestroy** ()
- virtual void **OnGrabbingJointBroken** ()

Invoked when [Grabber.GrabbingJoint](#) breaks. When overriding this careful to ensure that the 'GrabJointBroken' event is still invoked and (if desired) that any non-null 'Grabbing' object is released.

Properties

- [GrabbableObject](#) **Grabbing** [get]
 A reference to the object being grabbed by this hand, otherwise null if nothing being grabbed.
- Joint **GrabbingJoint** [get]
 A reference to the Joint being used to connect a Rigidbody-containing [GrabbableObject](#), otherwise null.
- Transform **AnchorTransform** [get]
 Returns the Transform grabbables are anchored to.
- Transform **GrabberPivot** [get]
 Returns the Transform that is used as the pivot point for the [Grabber](#). This is mostly used when positioning the [Grabber](#) on a GrabbableObjects 'MaintainGrabTransform' reference.
- Transform **GrabPoint** [get]
 A reference to the Transform for the object that was created at the grabber position when the grab occurred. Parented to the grabbed object, this means 'GrabPoint' maintains a reference of the hands relative position relative to the grabbed object..
- bool **IsGrabbingNonKinematicRigidbody** [get]
 Returns true if the [Grabber](#) is currently grabbing a non-kinematic Rigidbody, otherwise false.
- int **JointCount** [get]
 Returns the number of joints that are created by this [Grabber](#) (No guarantee that some joint entries will not be null if they are destroyed without using [Grabber.DestroyJoint](#)).
- bool **CanRelease** [get]
 Returns true if this [Grabber](#) may release the object it is 'Grabbing', otherwise false. The 'grabLocked' boolean or [SetGrabLocked\(bool\)](#) method may be used to prevent a release. NOTE: The 'CanReleaseDelegate' event may be used to override release permissions.

Events

- ActionRef< [Grabber](#), [GrabbableObject](#), bool > **CanGrabDelegate**
 A simple event that is passed a boolean by reference that may be overridden to override whether or not a grab can occur.
- ActionRef< [Grabber](#), [GrabbableObject](#), bool > **CanReleaseDelegate**
 A simple event that is passed a boolean by reference that may be overridden to override whether or not a non-forced release can occur.

5.16.1 Detailed Description

The base component for all grabbers.

Author: Mathew Aloisio

5.16.2 Member Function Documentation

5.16.2.1 CanGrab()

```
virtual bool GrabSystem.Grabber.CanGrab (
    GrabbableObject pGrabbable ) [virtual]
```

Returns true if this [Grabber](#) can grab pGrabbable, otherwise false. NOTE: By default (when not overridden) it is guaranteed that this method takes into account 'CanGrabDelegate' event overrides.

Parameters

<i>pGrabbable</i>	
-------------------	--

Returns

true if this [Grabber](#) can grab pGrabbable, otherwise false.

5.16.2.2 CreateJoint()

```
Joint GrabSystem.Grabber.CreateJoint (
    GrabbableObject pGrabbable,
    Rigidbody pRigidbody,
    float pBreakForce,
    float pBreakTorque,
    ConfigurableJoint pGrabJointPrefab )
```

Creates a physics joint between a Rigidbody and the [Grabber](#).

Parameters

<i>pGrabbable</i>	The GrabbableObject being jointed to.
<i>pRigidbody</i>	The Rigidbody that is being jointed to the Grabber .
<i>pBreakForce</i>	The force it takes to break the joint.
<i>pBreakTorque</i>	The torque it takes to break the joint.
<i>pGrabJointPrefab</i>	The custom grab joint prefab to use. If null the default joint configuration will be used.

Returns

The Joint that was created.

5.16.2.3 DestroyJoint()

```
void GrabSystem.Grabber.DestroyJoint (
    Joint pJoint )
```

Destroys a Joint and removes it from this Grabbers tracked joint list if it is in it.

Parameters

<i>pJoint</i>	
---------------	--

5.16.2.4 GetGrabbableOffset()

```
Vector3 GrabSystem.Grabber.GetGrabbableOffset (
    GrabbableObject pGrabbable )
```

Returns the position of pGrabbable's transform in the local space of the [Grabber](#)'s transform.

Parameters

<i>pGrabbable</i>	
-------------------	--

Returns

the position of pGrabbable's transform in the local space of the [Grabber](#)'s transform.

5.16.2.5 GetGrabbableRotationOffset()

```
Quaternion GrabSystem.Grabber.GetGrabbableRotationOffset (
    GrabbableObject pGrabbable )
```

Returns the rotation offset of pGrabbable's transform relative to the [Grabber](#)'s transform.

Parameters

<i>pGrabbable</i>	
-------------------	--

Returns

the rotation offset of pGrabbable's transform relative to the [Grabber](#)'s transform.

5.16.2.6 GetJointAtIndex()

```
Joint GrabSystem.Grabber.GetJointAtIndex (
    int pIndex )
```

Returns the Joint at the given index of the joints array for this [Grabber](#).

Parameters

<i>pIndex</i>	
---------------	--

Returns

The Joint at the given index of the joints array for this [Grabber](#).

5.16.2.7 GetOffsetFromGrabber()

```
Vector3 GrabSystem.Grabber.GetOffsetFromGrabber (
    GrabbableObject pGrabbable )
```

Returns the offset of pGrabbable from 'this' [Grabber](#) as a Vector3 in local space of 'this' [Grabber](#).

Parameters

<i>pGrabbable</i>	The grabbable object.
-------------------	-----------------------

Returns

the offset of pGrabbable from 'this' [Grabber](#) as a Vector3 in local space of 'this' [Grabber](#).

5.16.2.8 GetRelativeRotationFromGrabber()

```
Quaternion GrabSystem.Grabber.GetRelativeRotationFromGrabber (
    GrabbableObject pGrabbable )
```

Returns the relative rotation of pGrabbable from 'this' [Grabber](#) as a Quaternion.

Parameters

<i>pGrabbable</i>	The grabbable object.
-------------------	-----------------------

Returns

the relative rotation of pGrabbable from 'this' [Grabber](#) as a Quaternion.

5.16.2.9 GetTransformOffset()

```
Vector3 GrabSystem.Grabber.GetTransformOffset (
    Transform pTransform )
```

Returns the position of pTransform in the local space of the [Grabber](#)'s transform.

Parameters

<i>pTransform</i>	
-------------------	--

Returns

the position of pTransform in the local space of the [Grabber](#)'s transform.

5.16.2.10 GetTransformRotationOffset()

```
Quaternion GrabSystem.Grabber.GetTransformRotationOffset (
    Transform pTransform )
```

Returns the rotation offset of pTransform relative to the [Grabber](#)'s transform.

Parameters

<i>pTransform</i>	
-------------------	--

Returns

the rotation offset of pTransform relative to the [Grabber](#)'s transform.

5.16.2.11 Grab() [1/2]

```
void GrabSystem.Grabber.Grab (
    GrabbableObject pGrabbable )
```


Forces a [GrabbableObject](#) to be grabbed, if an object is already grabbed it is (attempted to be) released. Same as calling 'Grab(pGrabbable, null)'.

Parameters

<i>pGrabbable</i>	
-------------------	--

5.16.2.12 Grab() [2/2]

```
void GrabSystem.Grabber.Grab (
    GrabbableObject pGrabbable,
    Collider pGrabCollider )
```

Forces a [GrabbableObject](#) to be grabbed, if an object is already being grabbed it is (attempted to be) released.

Parameters

<i>pGrabbable</i>	The Grabbable that was grabbed.
<i>pGrabCollider</i>	The Collider that was grabbed.

5.16.2.13 GrabAtOffset()

```
void GrabSystem.Grabber.GrabAtOffset (
    GrabbableObject pGrabbable,
    Vector3 pOffset,
    Quaternion pRotation )
```

If pGrabbable is a movable object, teleports it to pOffset and pRotation local space (angle) offsets relative to the [Grabber's](#) 'transform', then grabs.

Parameters

<i>pGrabbable</i>	The grabbable object.
<i>pOffset</i>	The local space offset.
<i>pRotation</i>	The local space rotation.

5.16.2.14 SetGrabLocked()

```
void GrabSystem.Grabber.SetGrabLocked (
    bool pLocked )
```

Sets the 'grabLocked' field of this component. Useful for use with Unity editor events.

Parameters

<i>pLocked</i>	
----------------	--

5.16.2.15 ThrowRigidbody()

```
void GrabSystem.Grabber.ThrowRigidbody (
    Rigidbody pRigidbody,
    float pThrowForceMultiplier )
```

Calculates and applies throw forces to the specified Rigidbody.

Parameters

<i>pRigidbody</i>	
<i>pThrowForceMultiplier</i>	A throw force multiplier to apply to the calculated throw force. Useful for scaling force for specific objects.

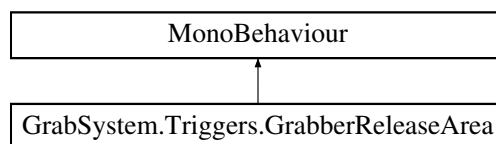
The documentation for this class was generated from the following file:

- Grabber.cs

5.17 GrabSystem.Triggers.GrabberReleaseArea Class Reference

A simple component that uses the 'OnTriggerEnter' and 'OnTriggerExit' callbacks to force a grabber to release on object when it enters the relevant trigger. Optionally prevents the hand from grabbing while in the release.

Inheritance diagram for GrabSystem.Triggers.GrabberReleaseArea:

**Public Types**

- enum **ReleaseMode**

Public Attributes

- ReleaseMode **releaseMode** = ReleaseMode.Release
The release mode to use when a [Grabber](#) enters this area.
InNone
- bool **forceRelease**
Should
- bool **checkRigidbody**
If not found on the triggering Collider
- bool **preventGrabbing**
Should grabbing be prevented for Grabbers in this area?
- [GrabberUnityEvent](#) **Triggered**
An event that is invoked when a [Grabber](#) triggers this component.
InArg
- [GrabUnityEvent](#) **GrabPrevented**
An event that is invoked whenever a [Grabber](#) is prevented from making a grab due to this component.
InArg

5.17.1 Detailed Description

A simple component that uses the 'OnTriggerEnter' and 'OnTriggerExit' callbacks to force a grabber to release on object when it enters the relevant trigger. Optionally prevents the hand from grabbing while in the release.

Author: Mathew Aloisio

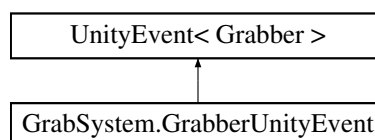
The documentation for this class was generated from the following file:

- GrabberReleaseArea.cs

5.18 GrabSystem.GrabberUnityEvent Class Reference

Arg0: [Grabber](#) - The [Grabber](#) involved in this grab event.

Inheritance diagram for GrabSystem.GrabberUnityEvent:



5.18.1 Detailed Description

Arg0: [Grabber](#) - The [Grabber](#) involved in this grab event.

The documentation for this class was generated from the following file:

- GrabberUnityEvent.cs

5.19 GrabSystem.Poses.MaintainOffsetByGrabOffset.GrabInfo Class Reference

Public Attributes

- [Entry](#) entry

A reference to the entry whose conditions were met.

The documentation for this class was generated from the following file:

- MaintainOffsetByGrabOffset.cs

5.20 GrabSystem.Poses.MaintainOffsetByProjectedAngle.GrabInfo Class Reference

Public Attributes

- [Entry](#) entry

A reference to the entry whose conditions were met.

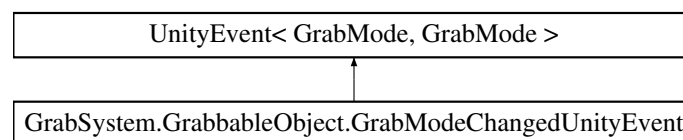
The documentation for this class was generated from the following file:

- MaintainOffsetByProjectedAngle.cs

5.21 GrabSystem.GrabbableObject.GrabModeChangedUnityEvent Class Reference

Arg0: GrabMode - The old grab mode. Arg1: GrabMode - The new grab mode.

Inheritance diagram for GrabSystem.GrabbableObject.GrabModeChangedUnityEvent:



5.21.1 Detailed Description

Arg0: GrabMode - The old grab mode. Arg1: GrabMode - The new grab mode.

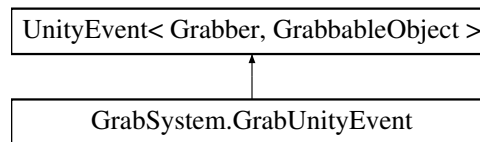
The documentation for this class was generated from the following file:

- GrabbableObject.cs

5.22 GrabSystem.GrabUnityEvent Class Reference

Arg0: [Grabber](#) - The [Grabber](#) involved in this grab event. Arg1: [GrabbableObject](#) - The [GrabbableObject](#) that was involved in the event.

Inheritance diagram for GrabSystem.GrabUnityEvent:



5.22.1 Detailed Description

Arg0: [Grabber](#) - The [Grabber](#) involved in this grab event. Arg1: [GrabbableObject](#) - The [GrabbableObject](#) that was involved in the event.

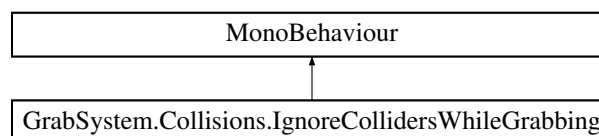
The documentation for this class was generated from the following file:

- `GrabUnityEvent.cs`

5.23 GrabSystem.Collisions.IgnoreCollidersWhileGrabbing Class Reference

A component intended to be attached to a [GrabbableObject](#) that subscribes to the relevant 'Grabbed' and 'Released' events to ignore collisions between some colliders and a [Grabber](#) while it is holding an object.

Inheritance diagram for GrabSystem.Collisions.IgnoreCollidersWhileGrabbing:



Public Attributes

- `Collider[] colliders`

An array of colliders to enable/disable collisions against the hand for.

- `ActionRef< Grabber, GrabbableObject, Collider[], bool > OverrideUnignoreDelegate`

A delegate event that allows the 'unignore collisions' behaviour to be overridden by anything that subscribes to this event. The boolean argument is passed as a reference, when flipped to true the original unignore collisions behaviour will be skipped. Arg0: [Grabber](#) - The grabber whose collisions are ignored with 'colliders'. Arg1: [GrabbableObject](#) - The grabbable object who is associated with the collision ignore event. Arg2: `Collider[]` - The array of Colliders the grabber is ignoring. Arg3: `ref bool` - The 'skip unignore collisions' boolean reference that when made true will make the 'unignore collisions' behaviour of this component get skipped.

Properties

- [GrabbableObject](#) **Grabbable** [get]

A reference to the [GrabbableObject](#) associated with this component.

5.23.1 Detailed Description

A component intended to be attached to a [GrabbableObject](#) that subscribes to the relevant 'Grabbed' and 'Released' events to ignore collisions between some colliders and a [Grabber](#) while it is holding an object.

NOTE: This component does not manage triggers. Author: Mathew Aloisio

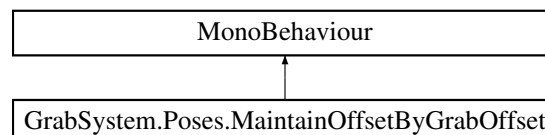
The documentation for this class was generated from the following file:

- IgnoreCollidersWhileGrabbing.cs

5.24 GrabSystem.Poses.MaintainOffsetByGrabOffset Class Reference

A component that is attached to the same GameObject as a [GrabbableObject](#) (that is assumed to be using the 'Maintain Offset' grab mode) which overrides the maintain offset behaviour allowing more complex behaviours to be defined using the grab offset of the [Grabber](#) from the grabbable at the time it grabbed the [GrabbableObject](#). NOTE: This component works in the grabbables local space by comparing the 'signed distance in direction' of the grabber from the grabbable at the time of grab using the comparator specified in the entry.

Inheritance diagram for GrabSystem.Poses.MaintainOffsetByGrabOffset:



Classes

- class [Entry](#)
- class [GrabInfo](#)

Public Member Functions

- bool [CheckEntryConditions](#) ([Entry](#) pEntry, [Grabber](#) pGrabber, [GrabbableObject](#) pGrabbable)
A public method that checks if the conditions are met for the given entry, pEntry, based on the relative offset of pGrabber from pGrabbable in pGrabbable local space.

Public Attributes

- [Entry\[\]](#) **entries**
An array of MaintainOffsetByGrabOffset.Entrys that contain rules that decide which grab pose is used.

Properties

- [GrabbableObject](#) **Grabbable** [get]

Returns the [GrabbableObject](#) reference associated with this component.

5.24.1 Detailed Description

A component that is attached to the same GameObject as a [GrabbableObject](#) (that is assumed to be using the 'Maintain Offset' grab mode) which overrides the maintain offset behaviour allowing more complex behaviours to be defined using the grab offset of the [Grabber](#) from the grabbable at the time it grabbed the [GrabbableObject](#). NOTE: This component works in the grabbables local space by comparing the 'signed distance in direction' of the grabber from the grabbable at the time of grab using the comparator specified in the entry.

Author: Intuitive Gaming Solutions

5.24.2 Member Function Documentation

5.24.2.1 CheckEntryConditions()

```
bool GrabSystem.Poses.MaintainOffsetByGrabOffset.CheckEntryConditions (
    Entry pEntry,
    Grabber pGrabber,
    GrabbableObject pGrabbable )
```

A public method that checks if the conditions are met for the given entry, pEntry, based on the relative offset of pGrabber from pGrabbable in pGrabbable local space.

Parameters

<i>pEntry</i>	
<i>pGrabber</i>	
<i>pGrabbable</i>	

Returns

true if the conditions for this entry are met, otherwise false.

The documentation for this class was generated from the following file:

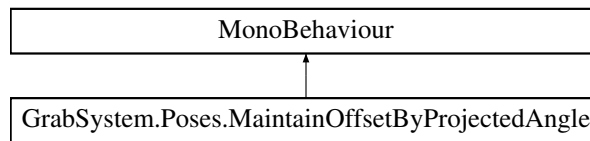
- MaintainOffsetByGrabOffset.cs

5.25 GrabSystem.Poses.MaintainOffsetByProjectedAngle Class Reference

A component that is attached to the same GameObject as a [GrabbableObject](#) (that is assumed to be using the 'Maintain Offset' grab mode) which overrides the maintain offset behaviour allowing more complex behaviours to

be defined using the relative angle of the [GrabbableObject](#) from the [Grabber](#) at the time it is grabbed. NOTE: This component works by projecting the 'grabber' onto the local 'axis' in local space of the grabbable and getting the hands angle around that axis.

Inheritance diagram for `GrabSystem.Poses.MaintainOffsetByProjectedAngle`:



Classes

- class [Entry](#)
- class [GrabInfo](#)

Public Member Functions

- bool [CheckEntryConditions](#) ([Entry](#) pEntry, [Grabber](#) pGrabber, [GrabbableObject](#) pGrabbable)

A public method that checks if the conditions are met for the given entry, pEntry, based on the relative offset of pGrabber from pGrabbable in pGrabbable local space.

Public Attributes

- [Entry\[\]](#) **entries**

An array of [MaintainOffsetRelativeAngle](#).Entries that contain rules that decide which grab pose is used.

Properties

- [GrabbableObject](#) **Grabbable** [get]

Returns the [GrabbableObject](#) reference associated with this component.

5.25.1 Detailed Description

A component that is attached to the same GameObject as a [GrabbableObject](#) (that is assumed to be using the 'Maintain Offset' grab mode) which overrides the maintain offset behaviour allowing more complex behaviours to be defined using the relative angle of the [GrabbableObject](#) from the [Grabber](#) at the time it is grabbed. NOTE: This component works by projecting the 'grabber' onto the local 'axis' in local space of the grabbable and getting the hands angle around that axis.

Author: Intuitive Gaming Solutions

5.25.2 Member Function Documentation

5.25.2.1 CheckEntryConditions()

```
bool GrabSystem.Poses.MaintainOffsetByProjectedAngle.CheckEntryConditions (
    Entry pEntry,
    Grabber pGrabber,
    GrabbableObject pGrabbable )
```

A public method that checks if the conditions are met for the given entry, pEntry, based on the relative offset of pGrabber from pGrabbable in pGrabbable local space.

Parameters

<i>pEntry</i>	
<i>pGrabber</i>	
<i>pGrabbable</i>	

Returns

true if the conditions for this entry are met, otherwise false.

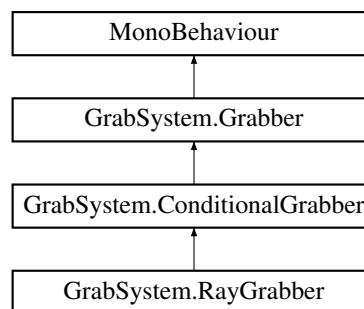
The documentation for this class was generated from the following file:

- MaintainOffsetByProjectedAngle.cs

5.26 GrabSystem.RayGrabber Class Reference

A component that extends a [Grabber](#) by adding the ability to attempt grabs using a raycast.

Inheritance diagram for GrabSystem.RayGrabber:



Classes

- class [BoxCastData](#)
- struct [GrabbableEntry](#)

Public Member Functions

- [GrabbableObject GrabByPalmTrace](#) ()
Attempts to grab any nearby objects using a ray trace from the palm in the physics scene returned by calling `PhysicsSceneUtility.GetLocalPhysicsScene(gameObject)`. Returns the [GrabbableObject](#) that was grabbed or null if nothing was grabbed.
- [GrabbableObject GrabByPalmTrace](#) (PhysicsScene pPhysicsScene)
Attempts to grab any nearby objects using a ray trace from the palm in the specified PhysicsScene. Returns the [GrabbableObject](#) that was grabbed or null if nothing was grabbed.
- override [GrabbableObject ConditionalGrab](#) (PhysicsScene pPhysicsScene)
Attempts to grab using conditional testing in the specified physics scene.

Static Public Member Functions

- static void [DrawBoxCastDataGizmo](#) ([BoxCastData](#) pBoxCastData, Color pColor)
Draws the given box cast data. Uses pBoxCastData.origin to determine position, rotation, and direction.
- static void [DrawBoxCastDataGizmoAt](#) ([BoxCastData](#) pBoxCastData, Color pColor, Vector3 pPosition, Quaternion pRotation, Vector3 pDirection)
Draws the given box cast data at the given position and rotation in the given direction.

Public Attributes

- LayerMask **ignoreGrabLayers**
A LayerMask of layers the grab system should ignore.
- [BoxCastData](#) **boxCastSettings** = new [BoxCastData](#)() { origin = null, boxHeight = 0.04f, boxWidth = 0.01f, boxLength = 0.01f }
The settings that control the grab

Additional Inherited Members

5.26.1 Detailed Description

A component that extends a [Grabber](#) by adding the ability to attempt grabs using a raycast.

Author: Mathew Aloisio

5.26.2 Member Function Documentation

5.26.2.1 ConditionalGrab()

```
override GrabbableObject GrabSystem.RayGrabber.ConditionalGrab (
    PhysicsScene pPhysicsScene ) [virtual]
```

Attempts to grab using conditional testing in the specified physics scene.

Parameters

<i>pPhysicsScene</i>	
----------------------	--

Returns

The [GrabbableObject](#) that was grabbed, otherwise null.

Implements [GrabSystem.ConditionalGrabber](#).

5.26.2.2 DrawBoxCastDataGizmo()

```
static void GrabSystem.RayGrabber.DrawBoxCastDataGizmo (
    BoxCastData pBoxCastData,
    Color pColor ) [static]
```

Draws the given box cast data. Uses pBoxCastData.origin to determine position, rotation, and direction.

Parameters

<i>pBoxCastData</i>	
<i>pColor</i>	The color to draw the gizmo in.

5.26.2.3 DrawBoxCastDataGizmoAt()

```
static void GrabSystem.RayGrabber.DrawBoxCastDataGizmoAt (
    BoxCastData pBoxCastData,
    Color pColor,
    Vector3 pPosition,
    Quaternion pRotation,
    Vector3 pDirection ) [static]
```

Draws the given box cast data at the given position and rotation in the given direction.

Parameters

<i>pBoxCastData</i>	
<i>pColor</i>	The color to draw the gizmo in.
<i>pPosition</i>	The position to draw the box cast at.
<i>pRotation</i>	The rotation to draw the box cast with.
<i>pDirection</i>	The box cast direction.

5.26.2.4 GrabByPalmTrace() [1/2]

```
GrabbableObject GrabSystem.RayGrabber.GrabByPalmTrace ( )
```

Attempts to grab any nearby objects using a ray trace from the palm in the physics scene returned by calling PhysicsSceneUtility.GetLocalPhysicsScene(gameObject). Returns the [GrabbableObject](#) that was grabbed or null if nothing was grabbed.

Returns

A reference to the [GrabbableObject](#) that was grabbed, otherwise null if nothing.

5.26.2.5 GrabByPalmTrace() [2/2]

```
GrabbableObject GrabSystem.RayGrabber.GrabByPalmTrace (
    PhysicsScene pPhysicsScene )
```

Attempts to grab any nearby objects using a ray trace from the palm in the specified PhysicsScene. Returns the [GrabbableObject](#) that was grabbed or null if nothing was grabbed.

Parameters

<i>pPhysicsScene</i>	
----------------------	--

Returns

A reference to the [GrabbableObject](#) that was grabbed, otherwise null if nothing.

The documentation for this class was generated from the following file:

- RayGrabber.cs

5.27 GrabSystem.Collisions.DelayUnignoreCollidersOnRelease.↩ UnignoreEntry Struct Reference

Public Attributes

- [Grabber](#) **grabber**
The [Grabber](#) to unignore collisions with.
- [GrabbableObject](#) **grabbable**
The [GrabbableObject](#) that was associated with the collision ignore event.
- float **unignoreTime**
The Time.time to unignore the collision at.
- Collider[] **unignoreColliders**
The array of Colliders to unignore.

The documentation for this struct was generated from the following file:

- DelayUnignoreCollidersOnRelease.cs

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